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## Amazon CloudFront – Distribution Creation (Step-by-Step)

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### Objective:

Create a CloudFront distribution to serve content (e.g., from an S3 bucket or HTTP server) with global caching and low latency.

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### Prerequisites:

- An existing **S3 bucket** or **origin server** (e.g., EC2, ALB, or external web server)
  - Content uploaded in the origin (e.g., HTML, images)
  - Optional: A **custom domain** (e.g., via **Route 53**)
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### Steps to Create a CloudFront Distribution

#### 1. Open the CloudFront Console

 Go to: <https://console.aws.amazon.com/cloudfront/>

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#### 2. Click "Create Distribution"

You'll be asked to choose between:

- **Web** (HTTP/S delivery – standard)

- **RTMP** (deprecated – for streaming)

➡ Choose **"Web" distribution** (default).

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### 3. Configure Origin Settings

| Setting                     | Value                                                                             |
|-----------------------------|-----------------------------------------------------------------------------------|
| Origin Domain Name          | Select your S3 bucket or enter custom origin (e.g., <a href="#">example.com</a> ) |
| Origin Path                 | (Optional) Set subdirectory                                                       |
| Origin ID                   | Auto-filled or name your origin                                                   |
| Restrict Bucket Access      | <b>Yes</b> (recommended for private content)                                      |
| Origin Access Control (OAC) | Create new or use existing to allow CloudFront to access S3                       |

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### 4. Default Cache Behavior Settings

| Setting                           | Value                                             |
|-----------------------------------|---------------------------------------------------|
| Viewer Protocol Policy            | Redirect HTTP to HTTPS                            |
| Allowed HTTP Methods              | GET, HEAD (or include OPTIONS/PUT/POST as needed) |
| Cache and Origin Request Settings | Use recommended or customize headers, cookies     |
| Object Caching                    | Use origin cache headers or specify TTL           |
| Minimum TTL                       | Example: 0 (no caching)                           |
| Maximum TTL                       | Example: 31536000 (1 year)                        |

|                                             |                                                             |
|---------------------------------------------|-------------------------------------------------------------|
| <b>Compress Objects Automatically</b>       | Yes (Gzip/Brotli)                                           |
| <b>Lambda@Edge or Function associations</b> | (Optional) For request/response transformation associations |

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## 5. Distribution Settings

| Setting                                | Value                                                             |
|----------------------------------------|-------------------------------------------------------------------|
| <b>Price Class</b>                     | E.g., "Use only North America and Europe" or "All Edge Locations" |
| <b>Alternate Domain Names (CNAMEs)</b> | e.g., <a href="#">cdn.example.com</a> (Optional)                  |
| <b>Custom SSL Certificate</b>          | Use ACM-issued certificate if using custom domain                 |
| <b>Default Root Object</b>             | e.g., <a href="#">index.html</a>                                  |
| <b>Logging</b>                         | Enable if required (log to S3)                                    |
| <b>WAF Web ACL</b>                     | Attach if using AWS WAF for protection                            |
| <b>IPv6</b>                            | Enable (recommended)                                              |

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## 6. Review and Create

Click **“Create Distribution”**. The distribution will take a few minutes to deploy.

Once deployed:

- **CloudFront URL:** You’ll get a domain like:  
[d1234example.cloudfront.net](#)
- You can **use this domain** in your app or configure a **custom domain with Route 53**.

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### Optional Post-Setup Steps:

#### Restrict S3 Bucket Access (if using S3 origin)

- Attach **OAC** or use **Bucket Policy** to allow CloudFront access only.

#### Add Route 53 Alias Record

- Create an **A/AAAA Alias record** pointing your domain to CloudFront's domain.

#### Test TTL Settings

- Use `curl -I` to inspect **Cache-Control** headers.
- Simulate cache invalidation via CloudFront console or CLI.

#### Enable CloudFront Metrics

- Use CloudWatch to monitor request count, cache hit ratio, 4xx/5xx errors, etc.
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